Computational Statistics and NLP Lab

Website: csnlp.netlify.app

This Complete Data Science Roadmap covers Basic Python, Machine Learning, Deep Learning, and Natural Language Processing.

Python

- Python Variable
- Python data type
- Python data structure
  - List
  - Tuple
  - Set
  - String
  - Dictionary
- Faction
- Operator
  - Arithmetic operator
  - Relational operator
  - Logical operator
- Conditional Control Statement
  - IF statement
  - IF…. else statement
- Loop control statement
  - For loop
  - While loop
- User define function
- Lambda function
- Class Function
- Type Dictionary

***DSA

***NumPy

***Pandas

Machine Learning Algorithms

- Basic Concept:
  - Normalization
  - L1, L2 Normalization
  - Regression Accuracy measure
    - Mean Absolute Error
    - Mean Square Error (MSE)
    - Root Mean Square Error (RMSE)
    - Relative Absolute Error (RAE)
    - Relative Square Error (RSE)
  - Classification Accuracy measure
    - Jaccard Index
    - Precision
    - Recall
    - F1 score
    - Log loss
- Supervised Learning
  - Linear regression
  - Logistic regression
  - KNN
  - Decision Tree
  - Random Forest
  - Naive Bayes
  - SVM + Kernels
  - Gradient decent
  - Ensemble learning
  - Gradient Boosting (XGBoost, LightGBM, CatBoost)
  - Hyperparameter Tuning
  - What is K-fold Cross validation?
- Unsupervised Learning
  - K-means
  - Hierarchical clustering
  - Dimensionality reduction (PCA, ICA, LDA)

Deep Learning

- Neural Network
  - What is a neural network?
  - Forward Perceptron
  - Weight Initialization technique
  - Activation Function
    - Sigmoid
    - Tanh
    - ReLU
    - Leaky Relu
    - ELU
    - SoftMax
    - Softplus
    - Switch
  - Chain Rule
  - Backpropagation implementation
  - Batch Size
  - Epoch
  - Learning Rate
  - Learning Rate Decency
  - Convex and non-convex cost Function
  - Loss Function
    - Binary Cross Entropy Loss
    - Cross Entropy Loss
    - Hinge Loss
    - Categorical Cross Entropy Loss
    - Sparse Categorical Cross Entropy
    - SoftMax cross entropy loss
  - Optimization Rules
    - Gradient Decent
    - Different Type of Gradient Decent
    - Stochastic Gradient Decent
    - Mini Batch Gradient Decent
    - Gradient Decent with Momentum

- Advanced Optimizer
  - Adagrad
  - RMSProp
  - AdaDelta
  - Adam
  - NAdam
  - AMSGrad
- Vanishing Gradient Decent Problem
- Exploding Gradient Problem
- Gradient Clipping
- Single-layer NN in TensorFlow
- Multi-layer NN TensorFlow

- Convolutional Neural Networks (CNN)
- Recurrent Neural Networks (RNN)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Unit (GRU)
- Transformer Architecture
- Attention is All You Need
- LLM Learning Technique
  - Zero Shot
  - One Shot
  - Few Shot
- Encoder-only models
  - BERT
  - RoBERTa
  - ALBERT
- Decoder-only models (LLMs)
  - GPT
  - LLaMA
  - DeepSeek
  - Phi
- Encoder-decoder (Seq2Seq)
  - T5

- BART
- mT5
- Whisper (speech)

Natural Language Processing (NLP)

- Text Preprocessing
  - Tokenization
  - Lowercasing
  - Stop word
  - Stemming
  - Lemmatization

- Word Embeddings
  - Bag of Words
  - Count Vectorizer
  - TF-IDF vectorizer
  - N-grams
  - Distributional semantics
  - Word2Vec
  - AngWord2Vec
  - Glove
  - Fast Text

Additional

- **Fast API
- **Pydantic
- **Docker
- ** AWS, or, AZURE